



“BIOMASS mission Cal/Val activities - JENA”

On large reflector antennas as radar target for low-frequency synthetic aperture radar

7 min “BIOMASS mission Cal/Val activities”

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A last minute result from DLR, thanks A.Silva, J.Amao





Introduction

- Reflector antennas as accurate PEC structure, as target
  - Radio-telescope antennas, focal region details needed
  - Fixed pointing towards BIOMASS at max elevation (0-Doppler)
  - focal structure important
  - BIOMASS: No coverage in Europe: ITU
  - Good cooperation with University of Tasmania, AU
- Concluding remarks

University of Tasmania, Australia
Controls radio telescope antennas remotely

Lower left: 12m Shaped dual reflector
Right: 26m Prime focus reflector antenna

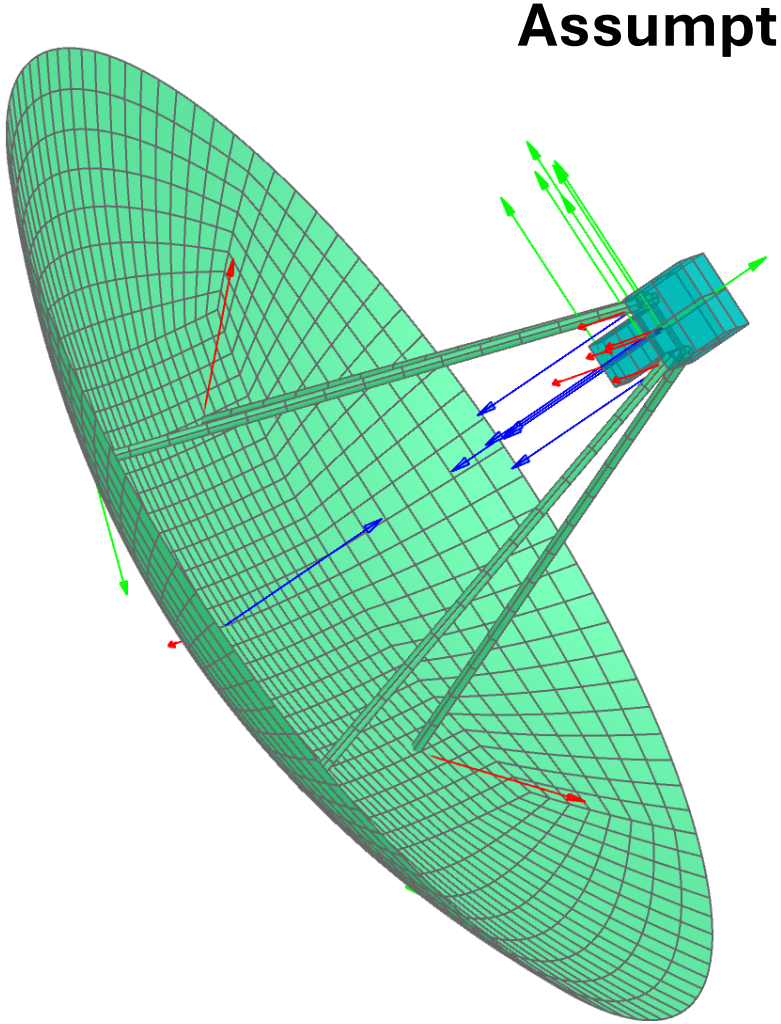
Several data sets collected with Biomass

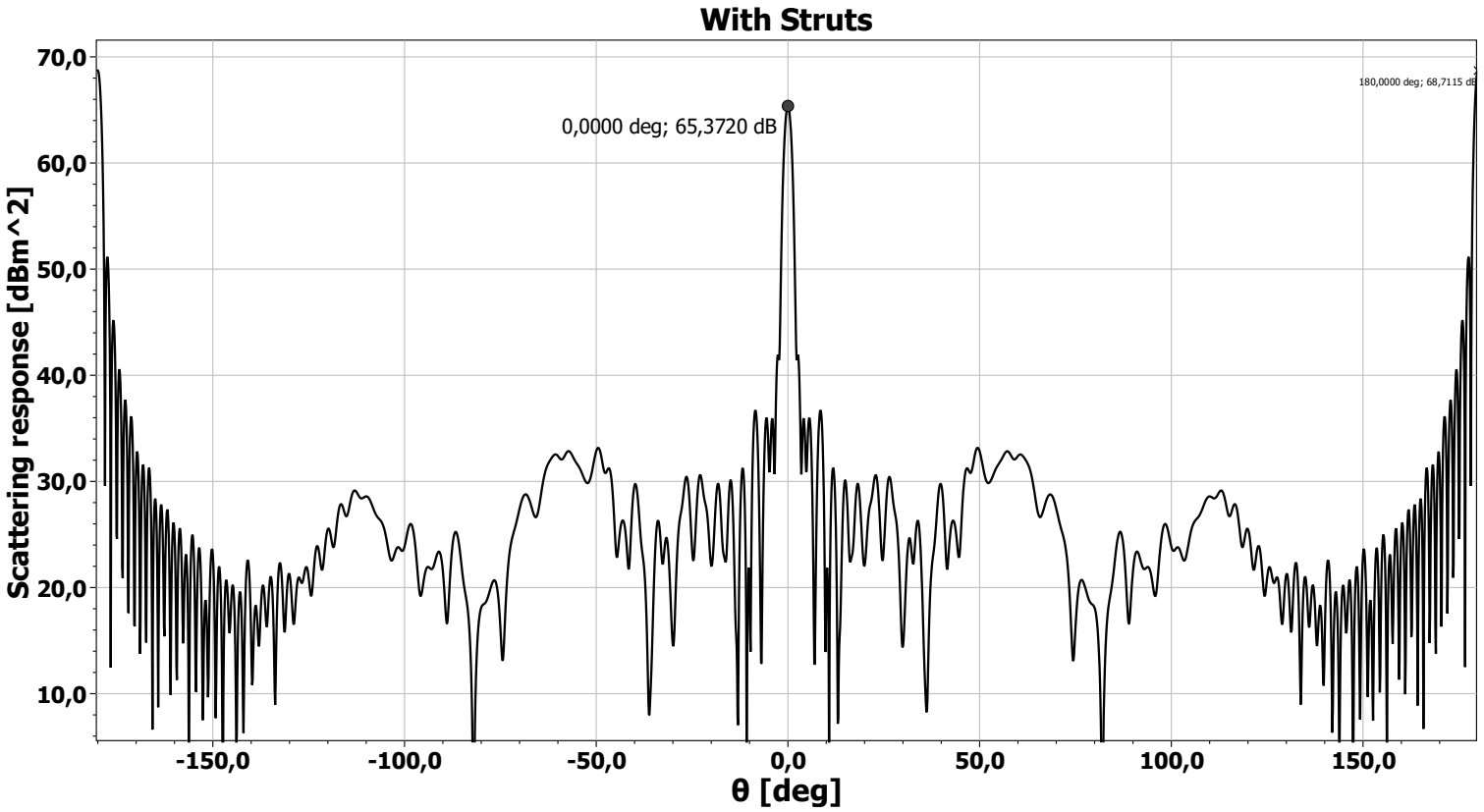
Copyright 2 fotos: University of Tasmania





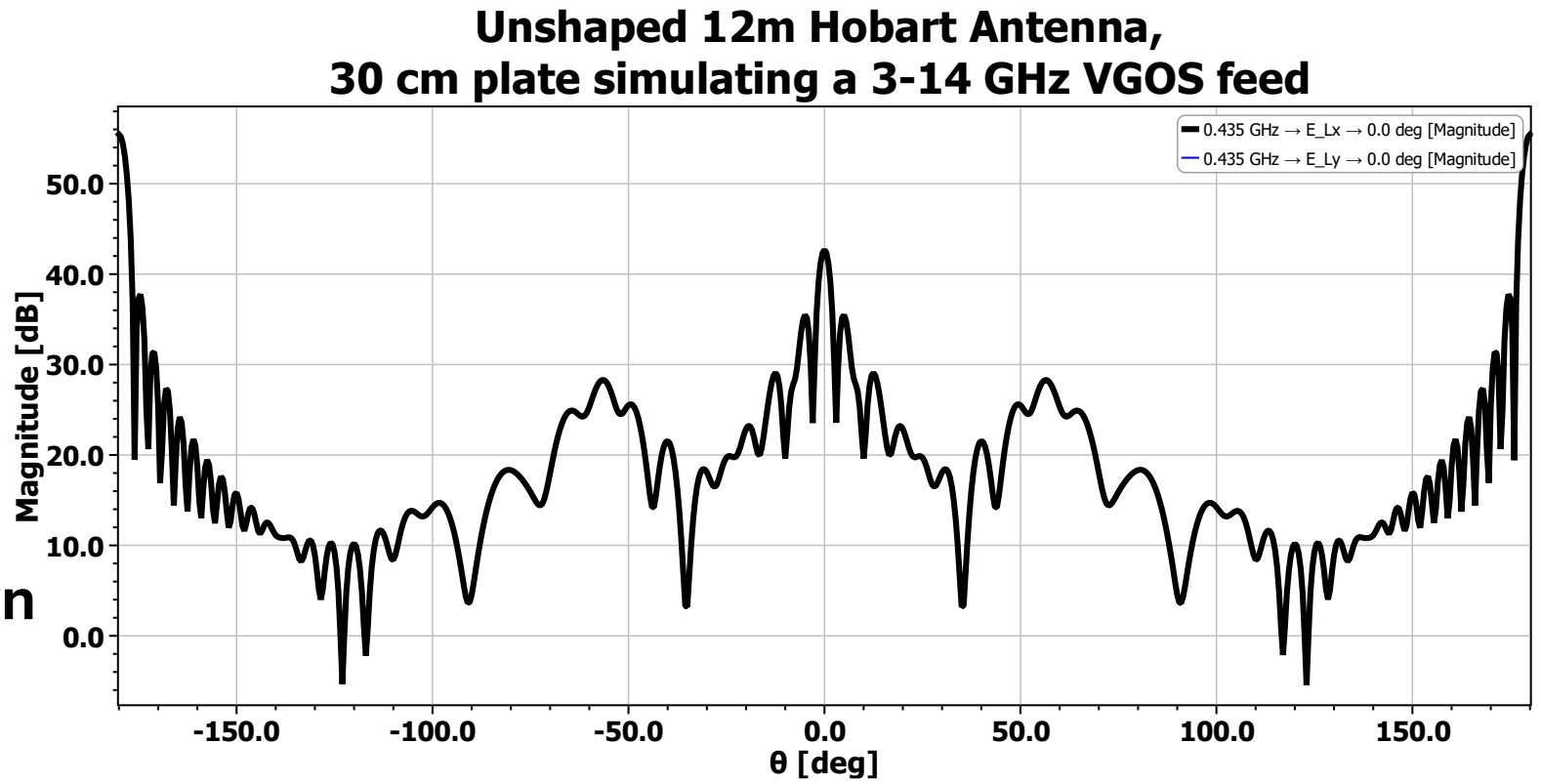
Preliminar prediction monostatic backscattering 26 m antenna ~ equivalent corner reflector would require a rib ~ 25m Assumptions for blockage focal container to be updated





Dual reflector 12m, Initial estimate MoM

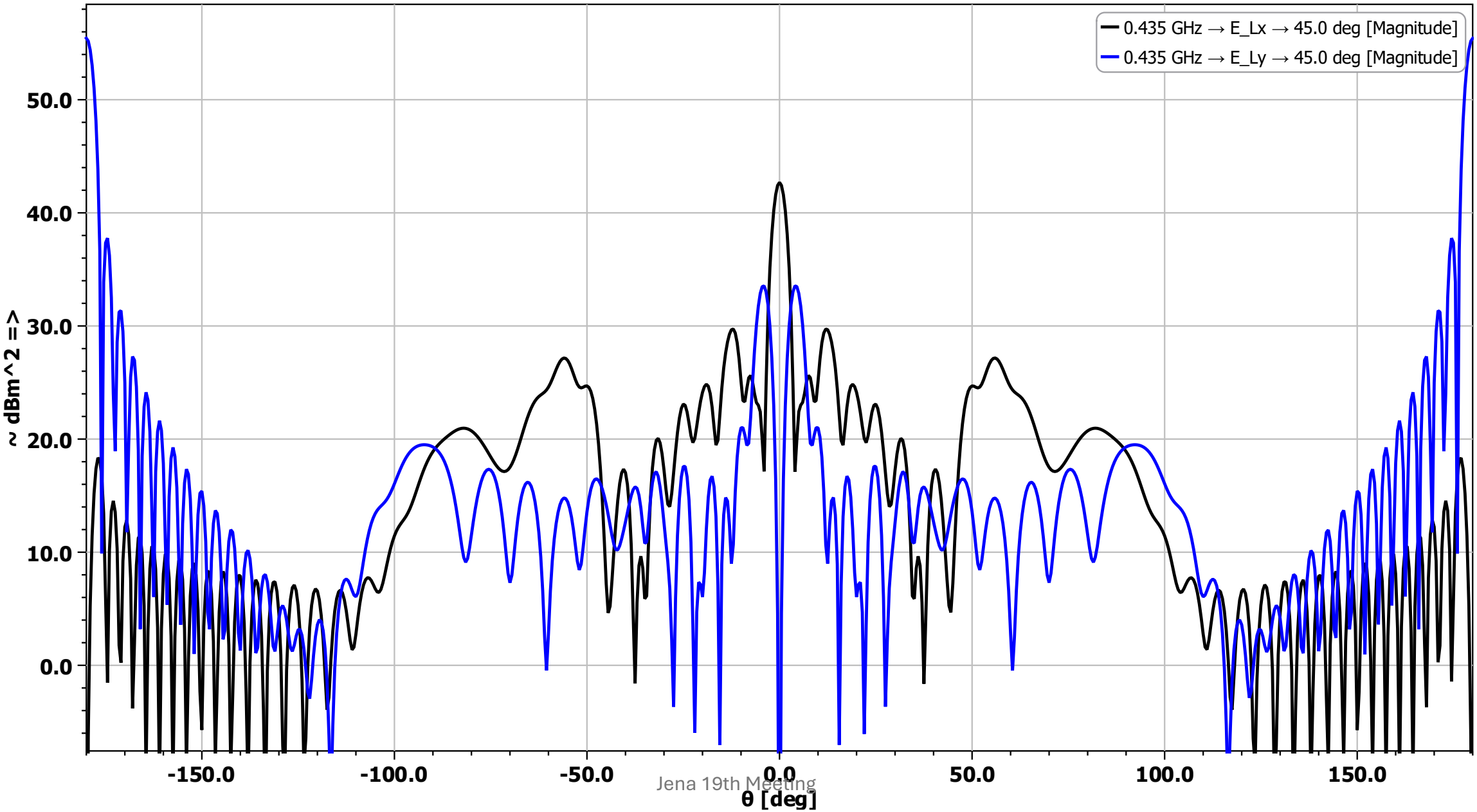
- No shaping
- No struts
- No feed tower
- Feed diam 30 cm
- Below cut-off in P-band
- MoM, full pol, PEC
- Co- and Cross-pol shown
- ~43 dBm^2 Max level



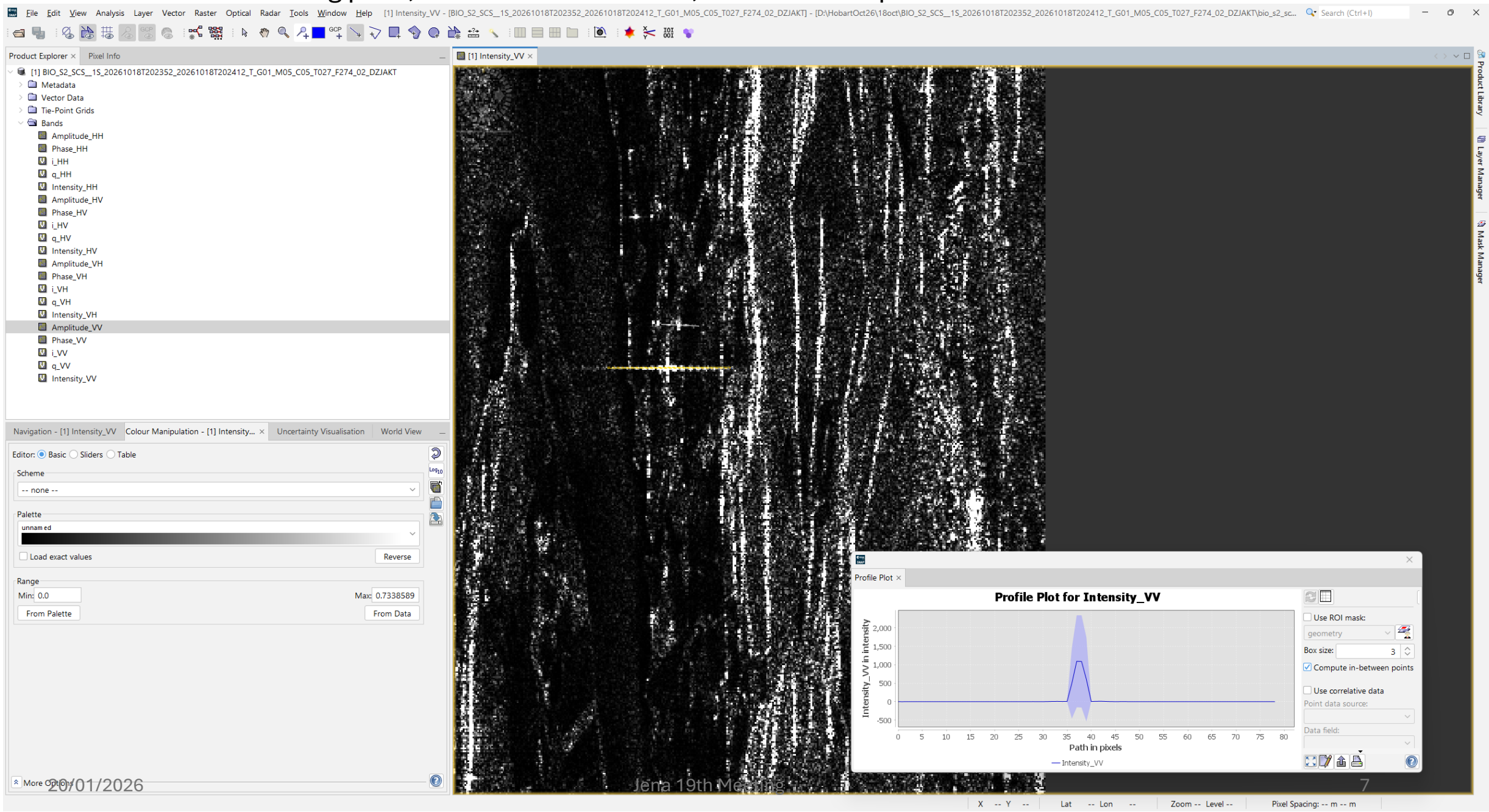


Xpol 45 deg plane 12m preliminary

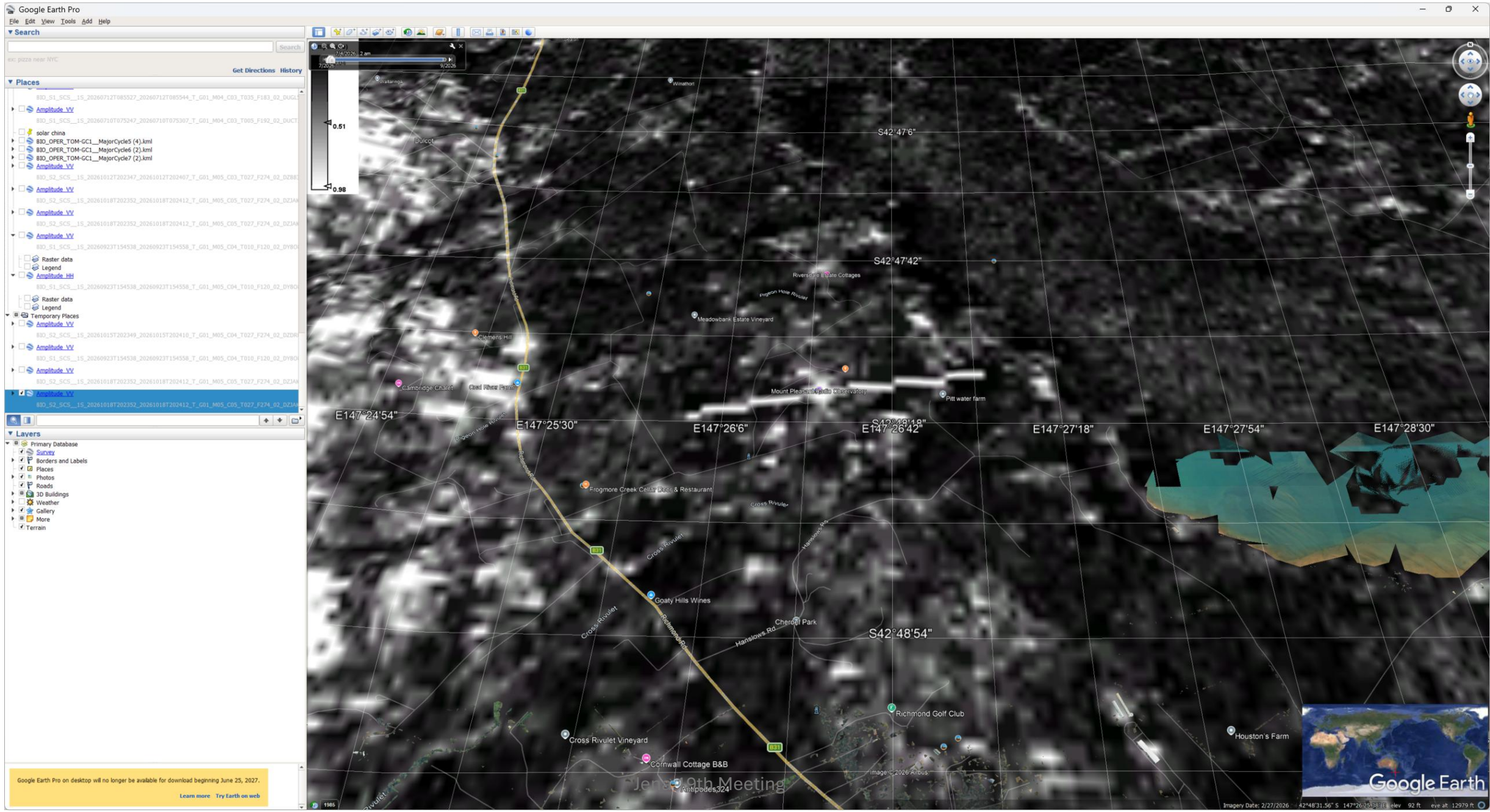
Monostatic back scattering, Co and Cross in 45 deg plane 12m , axial incidence



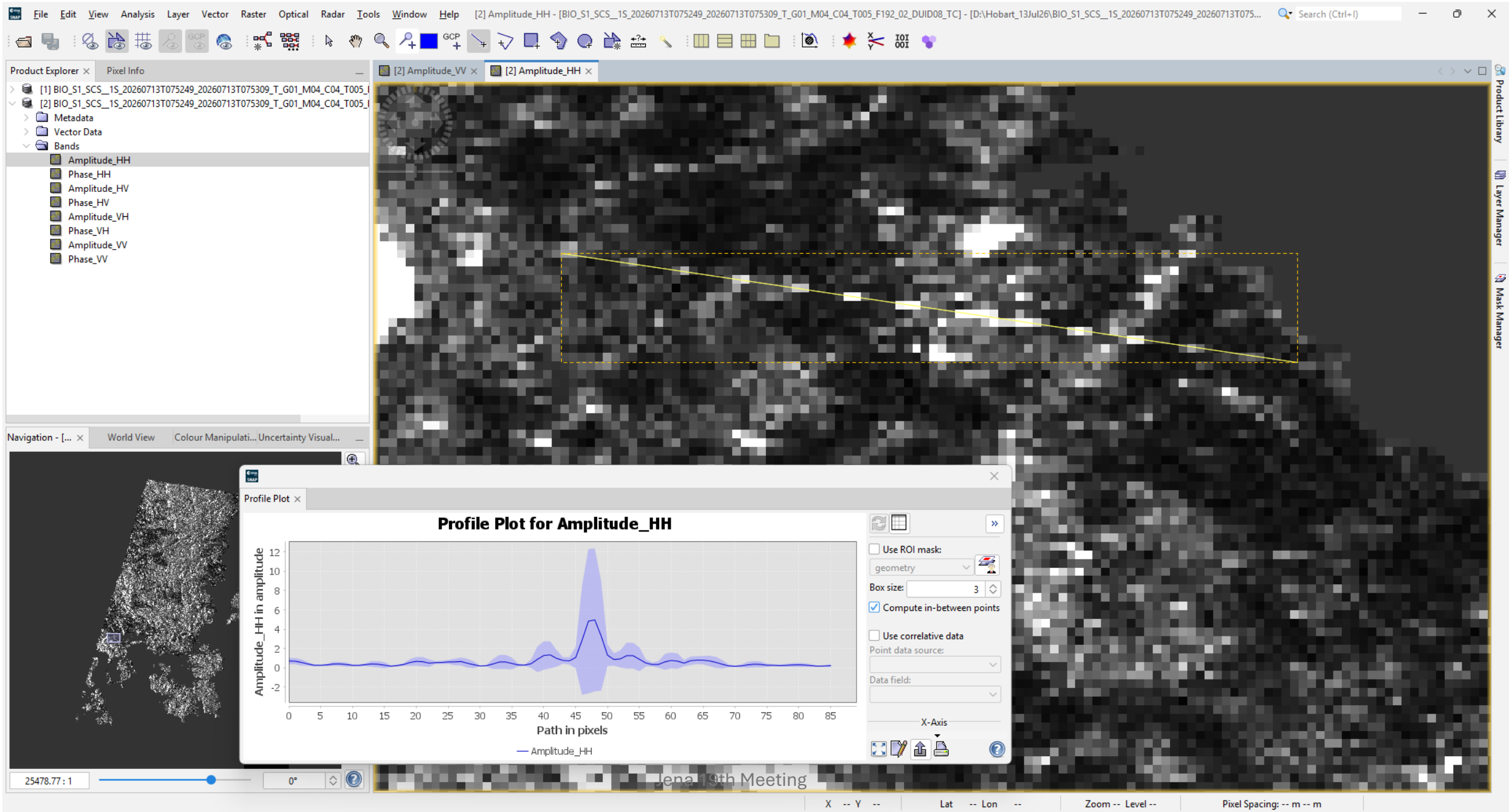
Recent ascending pass, 18 – 10 – 2026 Hobart, 26m antenna pointed



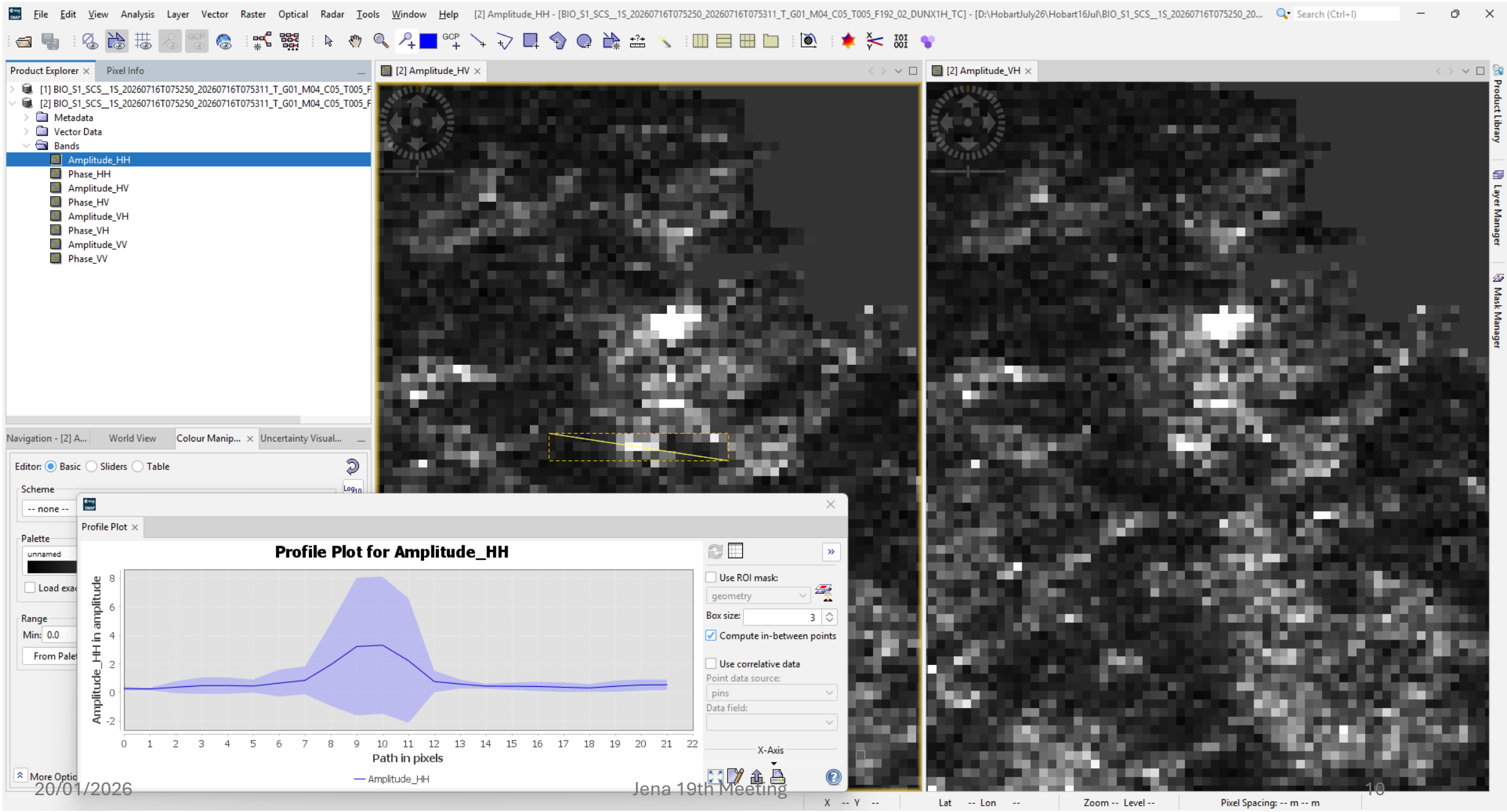
Recent Ascending pass, Hobart 26m

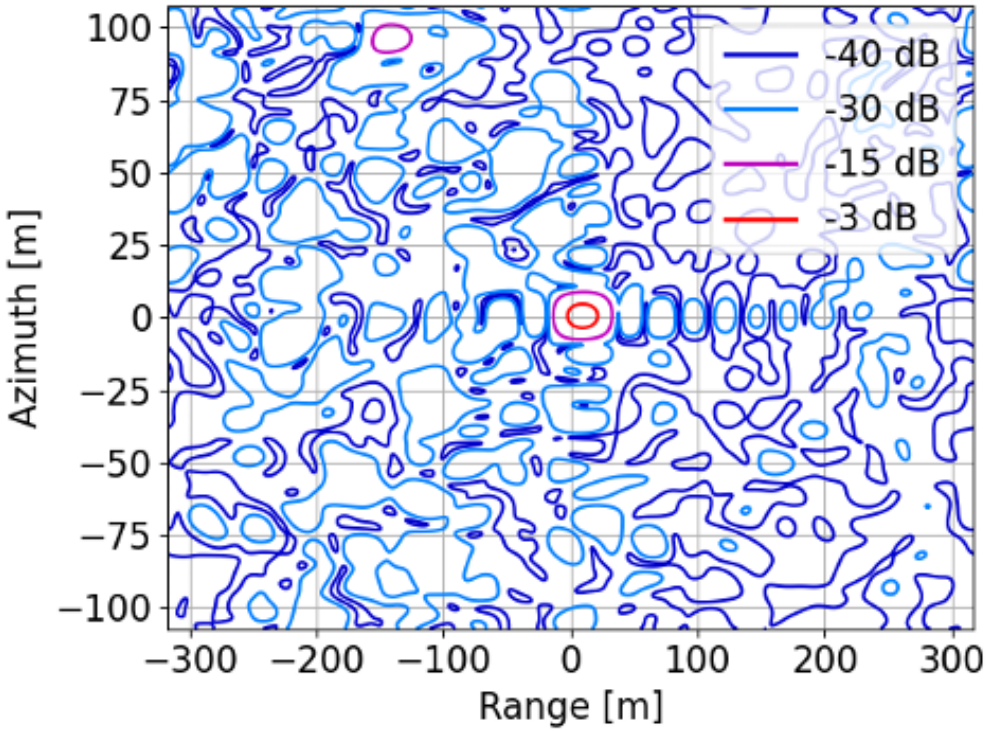


Descending pass, 13 July 2026, anticipated value too low.... Hobart 26m, but what is the absolute scale in SNAP13..

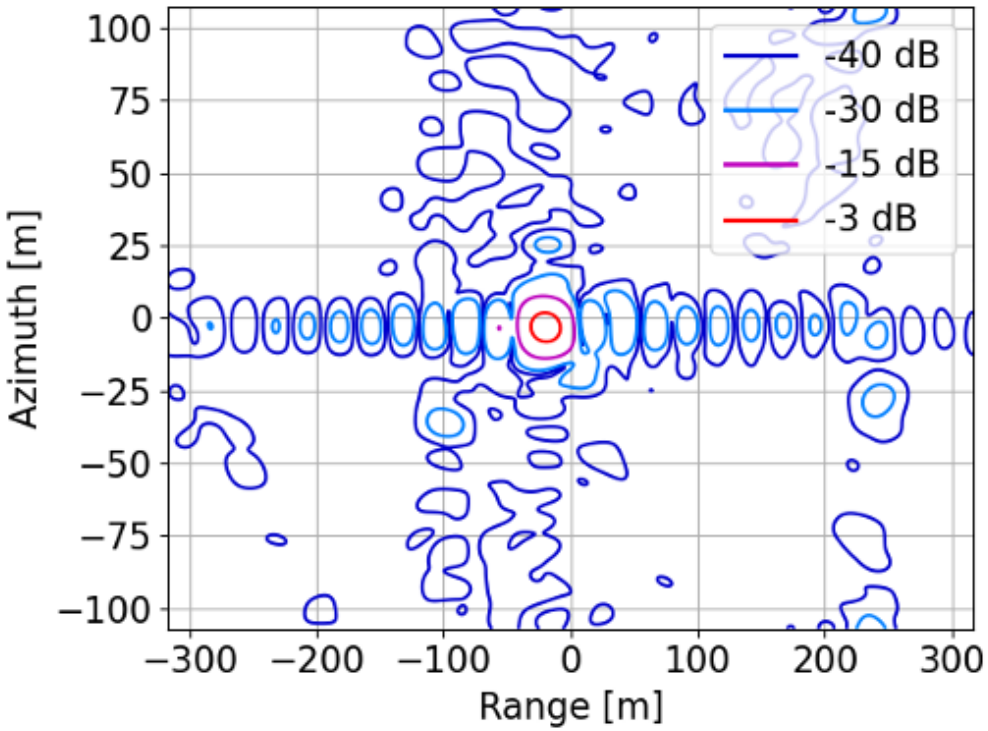


12m Hobart descending pass 16 July





Yarragadee 12meter 20-03-2026



Hobart 26meter 13-07-2025

Last minute contribution from DLR, - preliminary result – interesting!

➔Danke schőn and gracias and grazie

Further of interest: absolute peak level

➔Last minute preliminar data from DLR processing: Yarragadee 12m: 43.8658 dBm^2

➔ Hobart 26m: 61.564 dBm^2

Interesting: Hobart 12m and 26m antennas (separated by some 300m on Earth)➔interferometer

Concluding Remarks

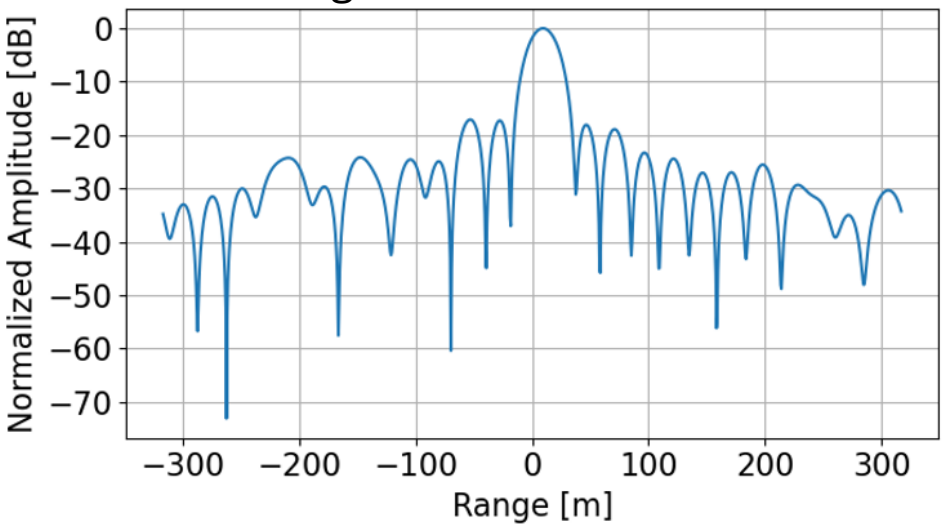
- Large reflector antennas as potential radar target indicative,
- Possible to predict full polarimetric scattering at 70 cm wavelength, but:
  - Detailed antenna structural information must be known,
    - Focal details, location focus, diameter feed aperture, feed assembly
  - But: There are many different type of reflector antennas.
    - Large reflector antennas are not directly available,
    - Large reflector antenna must be pointed to max elevation satellite (0-Doppler).
- Phase coherent scatterer useful, provided pointed,
- Further work needed for absolute level and polarization properties
  - Make model 12m and 26m antennas more precise, define range distance (θ_{inc} dependent)

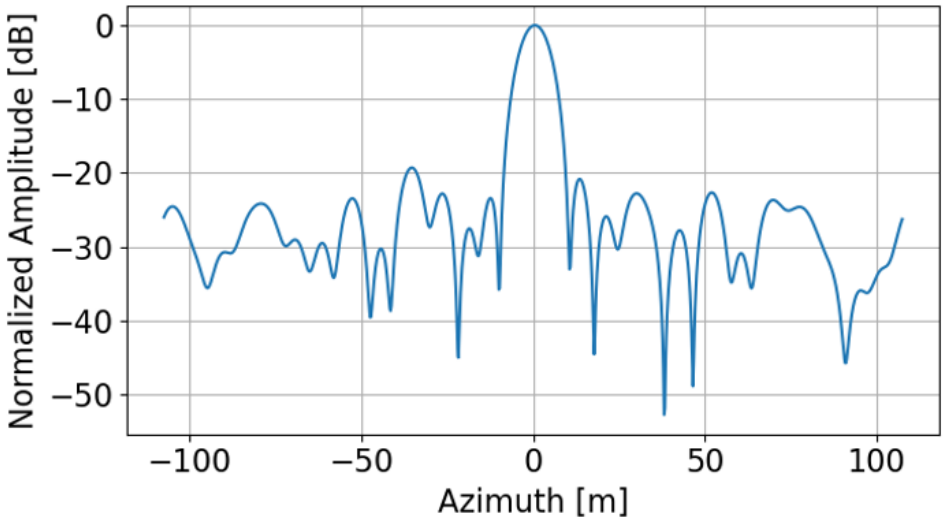
- ➔Exploit measurement and analysis in complement

- ➔Suggestion: inspect Westerbork Array as "phase-calibration bar"
  - With respect for ITU.....

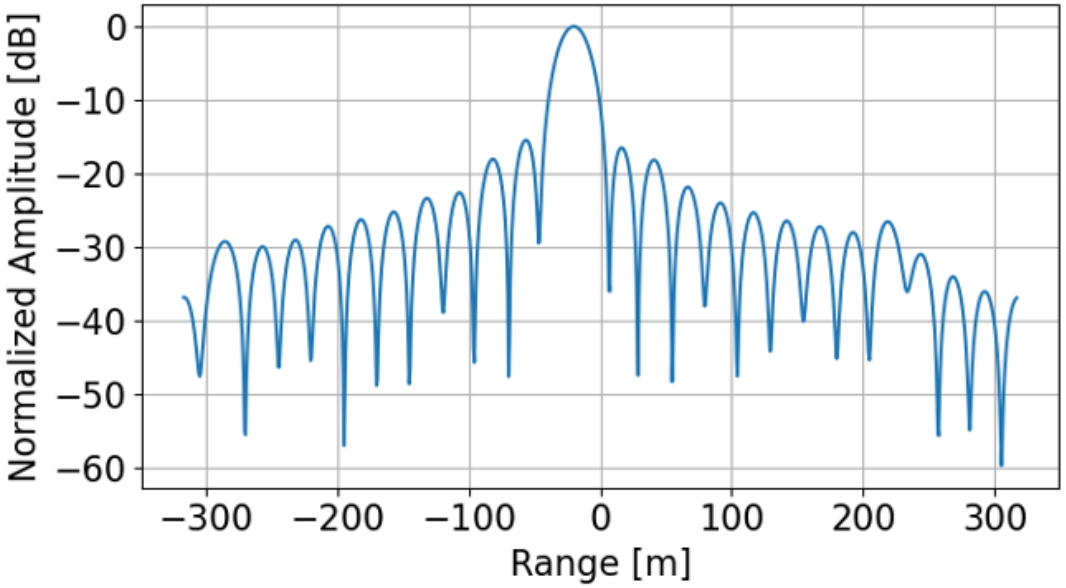
Interesting last minute preliminary results from SAR data (DLR)

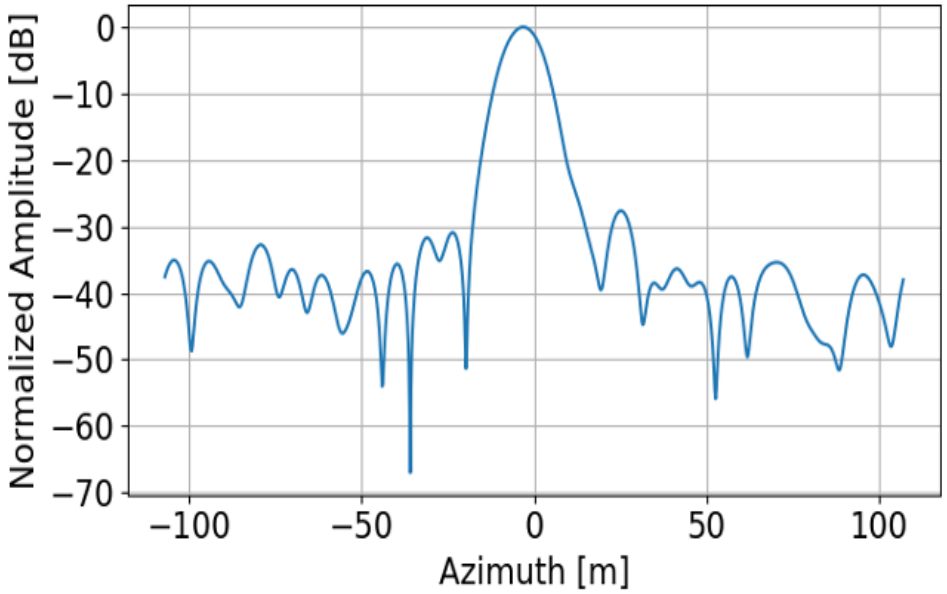
Yarragadee 12m





Hobart 26m

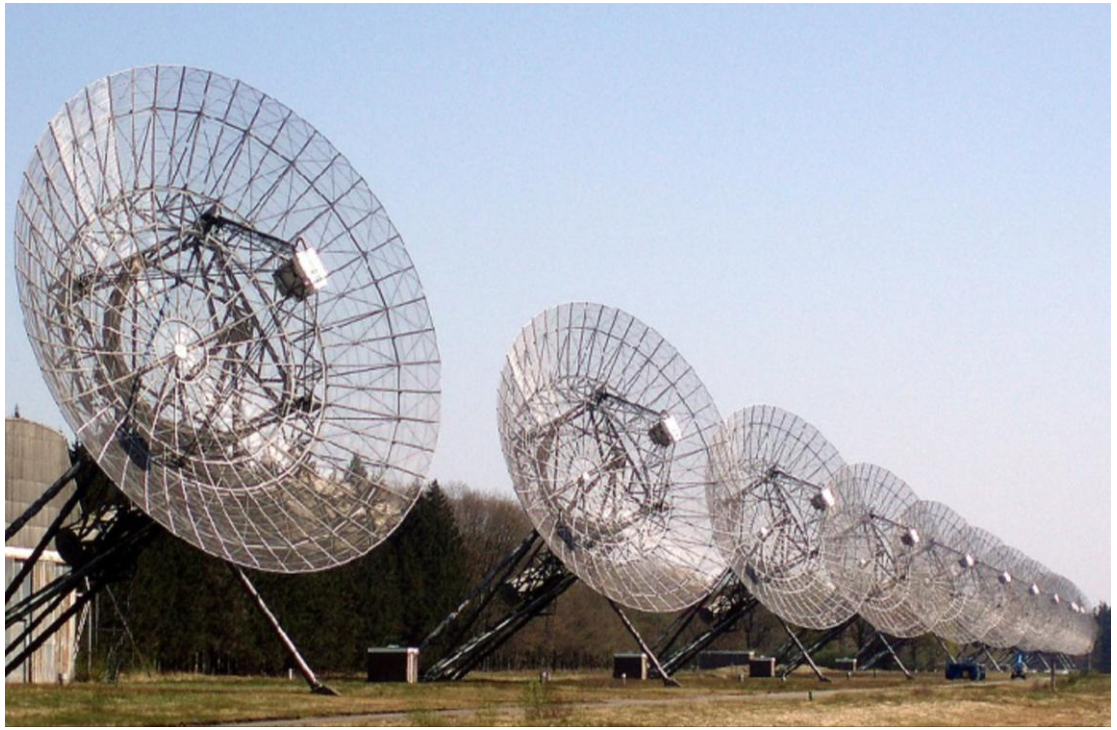


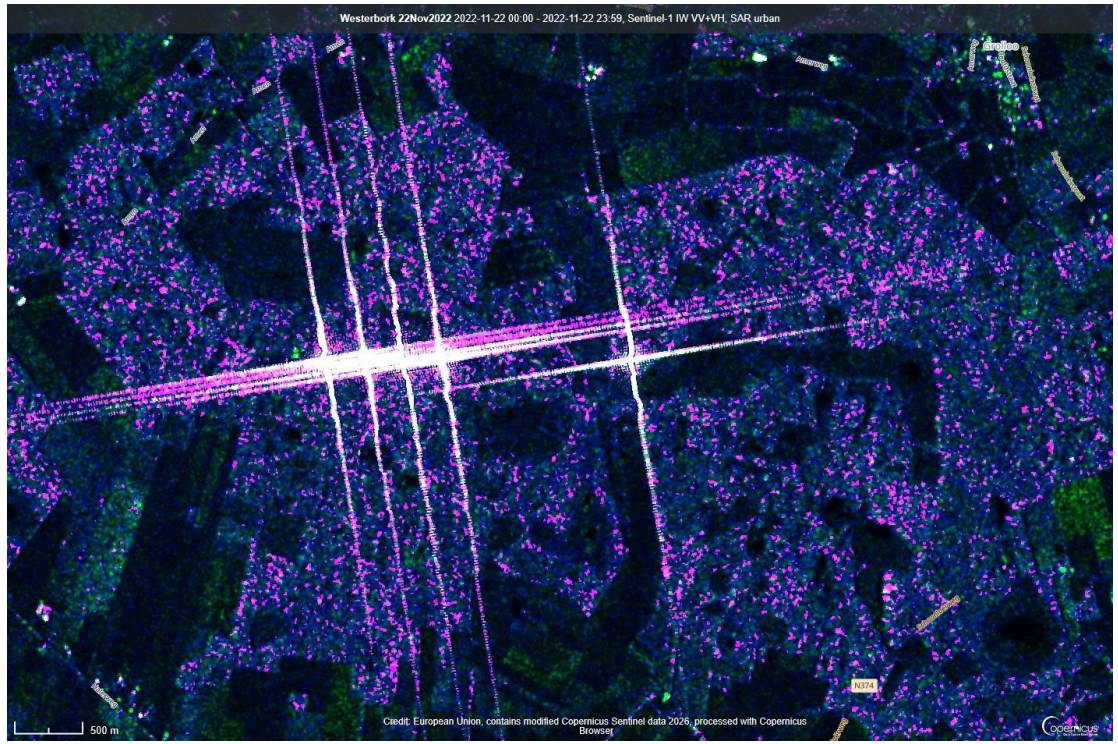


A few VG's for further info
Previous work, using Sentinel-1 data

Westerbork Radio Telescope Array, 25m antennas

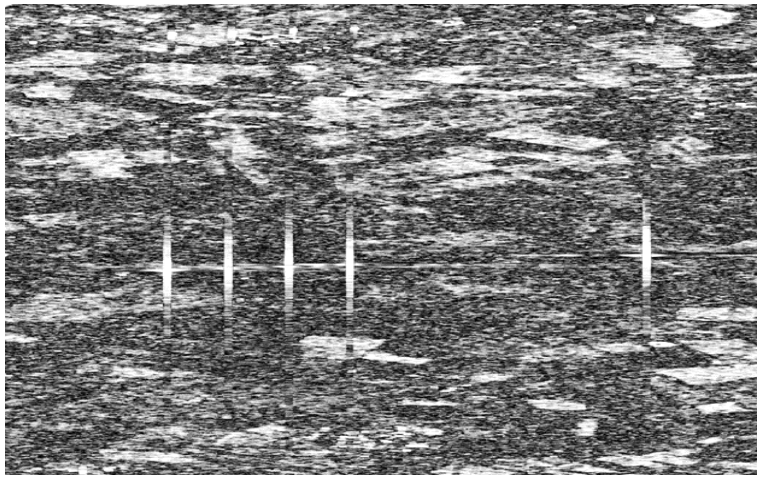
Five 25m antennas pointed → too strong level in C-band (below)
Of interest for BIOMASS, at 70 cm wavelength, as “calibration bar”
An array configuration of antennas each with comparable backscatter
2.8 km Length, East West, <u>Accuracy Focus Location: within +-1mm from baseline</u>



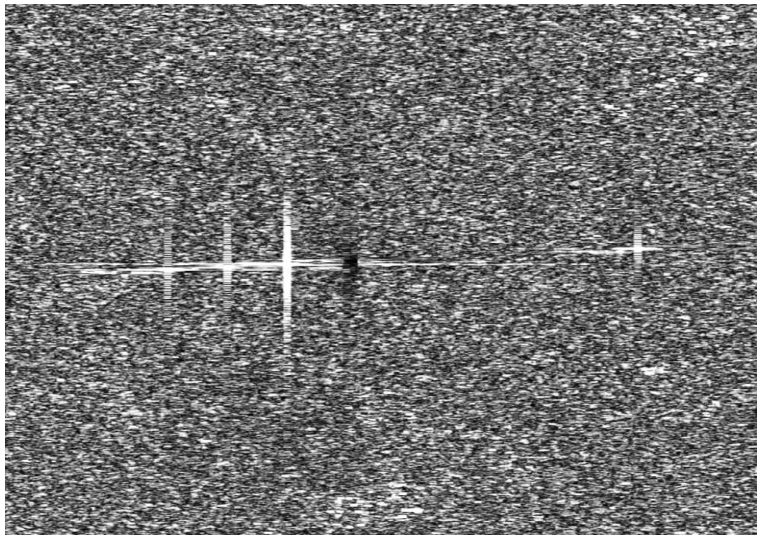


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Interferometry, Westerbork antennas as targets



5 Westerbork antennas pointed to Sentinel-1, 12 day interval, 5 antennas pointed Interferometry shows strong coherence,



5 Westerbork antennas pointed to Sentinel-1, 108 days interval,

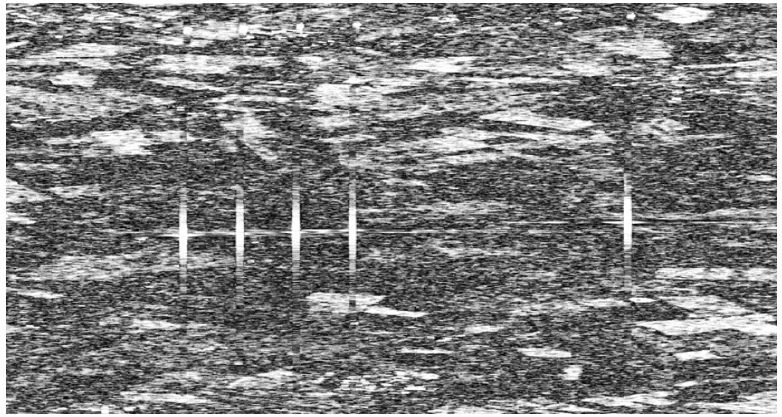
➔One antenna with a different pointing,

- **A strong incoherence noted (black)**

An incentive to explore interferometry

Interferometry with SNAP program, phase unwrapping

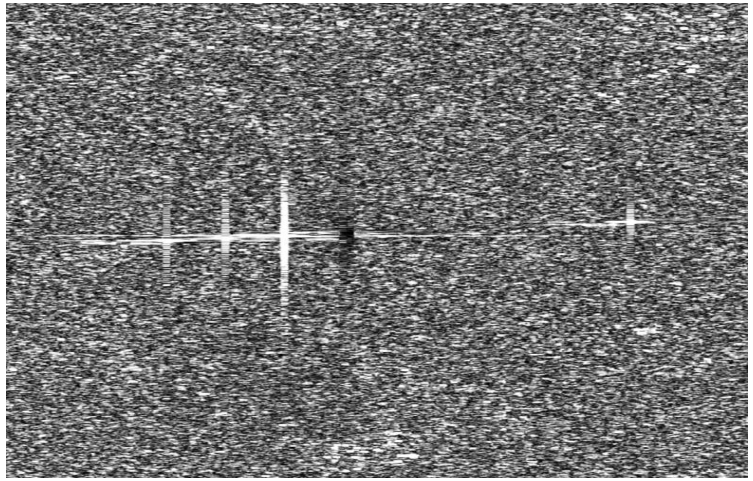
Coherence 12 days interval

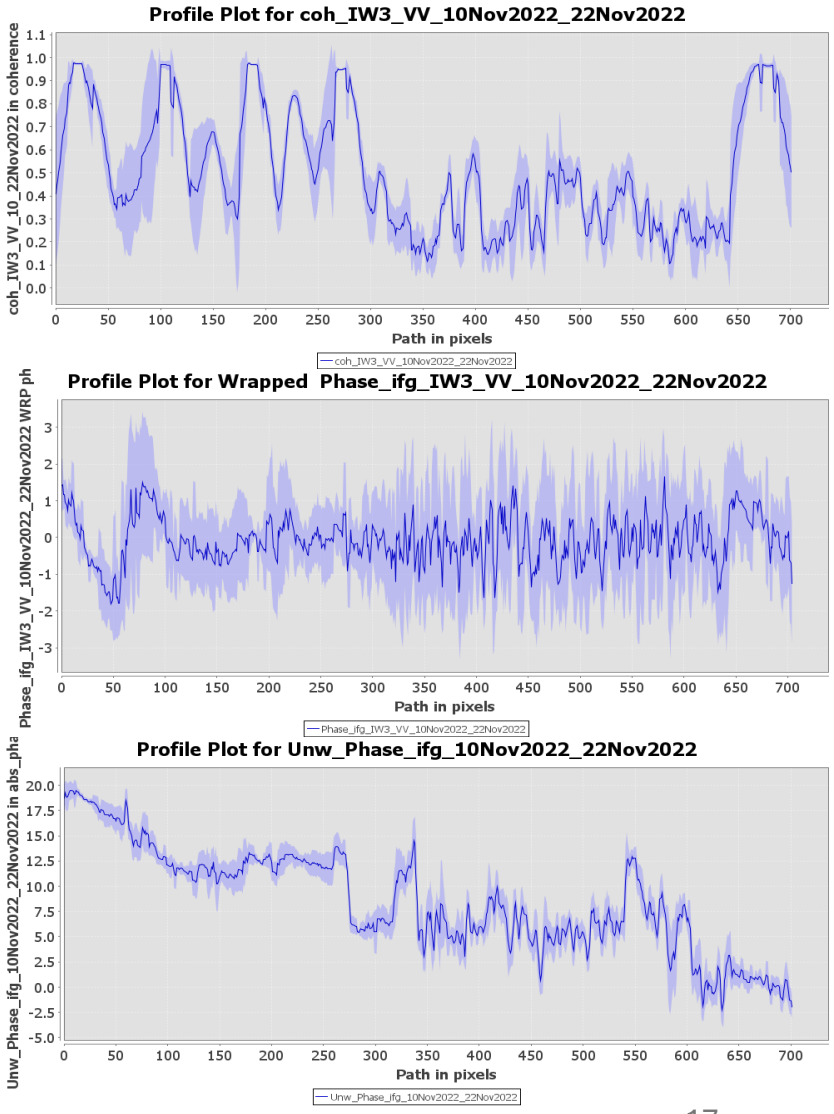


Pointing error of
Westerbork antenna observed
After 108 days from coherency

It invites for phase unwrapping
Despite strong response in
SLC data

Coherence 108 days interval





Dwingeloo 25m radio-telescope antenna

- Too strong signal for the 5.5 cm Sentinel SAR.
- Temporal modification in the focus ➔ ~2 orders change in monostatic backscatter -- Sentinel-1
- Explained in
- https://research.tue.nl/files/374210081/20260120_Klooster_van_t_hf.pdf



